

Tariq Aldalou

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EDUCATION

Georgia Institute of Technology — Atlanta, GA | M.S. Computer Engineering | Spring 2029

George Mason University — Fairfax, VA | B.S. Computer Engineering | **GPA: 3.95**

Dean's List: Fall 2024, Spring 2025, Fall 2025 | **Scholarships:** NVTC 2025, ADVANCED

Relevant Coursework: Computer Architecture (RISC-V, GPU/CPU/FPGA Design), RTL (VHDL), Embedded Systems, Internet of Things, ADA-SPARK, System Architecture Design, Microcontrollers, Operating System, Data Structures.

Northern Virginia Community College — Fairfax, VA | A.S. Engineering & Math | GPA: 3.90

Dean's List: Spring 2024, Fall 2022 | **Presidential Scholars:** Spring 2023, Fall 2023 | **Scholarships:** FAITH, Kimmy Duong

EXPERIENCE

Research Intern, AVATAR Research Program — Fairfax, VA

May 2026 – Present

- Contributing to AVATAR (Adaptive Virtual Assistant for Therapeutic Auto-Regulation), a multidisciplinary research program integrating computer engineering, bioengineering, and game design to build a VR and wearable biometric platform supporting addiction recovery
- Developing the wearable biometric sensing and signal-processing pipeline that captures physiological data to anticipate relapse-trigger states and deliver real-time recovery cues within an immersive VR environment

Quantum Computing Research Intern, Fermilab (Theory Division) — Batavia, IL

June 2025 – September 2025

- Completed intensive quantum computing research under Dr. Hank Lamm and Dr. Roni Harnik, mastering quantum gates, quantum coupling, and programming quantum circuits using Python and Qiskit SDK
- Executed quantum experiments on IBM 4-qubit superconducting quantum computers, comparing classical simulation results to real quantum hardware performance and analyzing big(O) complexity advantages

Engineering Program Instructor, Fairfax County Community Center — Fairfax, VA

April 2024 – Present

- Designed and delivered hands-on engineering curriculum for 12+ high school students across multiple 3-month cohorts, introducing circuit design, Arduino programming, and microcontroller fundamentals with 25% pursuing engineering fields
- Developed interactive STEM projects covering electronics theory, embedded systems programming, and emerging technologies, mentoring students in problem-solving and hardware prototyping techniques

PROJECTS

CubeSat FPGA Development Platform with Formal Verification | *Fall 2025 – Spring 2026*

- Contributing to team developing 3-layer PCB platform integrating Artix-7 FPGA (Basys3), power management, and modular sensor interfaces for formally verified and university-wide ADA-SPARK firmware development
- Designed complete power distribution layer with integrated power protection and temperature monitoring circuits; PCB final design ready for fabrication and stress-testing
- Developing formally verified I2C and SPI communication libraries in ADA-SPARK with modified UART registers for custom firmware upload, collaborating with AdaCore engineers on bootloader integration and safety-critical design

DriveGuard: IoT Driver Safety Monitoring System | *Fall 2025*

- Developed embedded firmware in C++ for Arduino Nano 33IoT integrating GPS(BN220) and IMU(LSM6DS3) sensors to detect unsafe driving patterns including excessive speed and harsh acceleration events
- Engineered local data buffering system storing 10 minutes of sensor reading, transmitting safety alerts via MQTT protocol with sub-2-seconds response time
- Designed complete hardware integration including sensor wiring, power management, real-time processing, PCB ready for fabrication, and real-time data processing on embedded microcontroller platform

TECHNICAL SKILLS

Languages: Python, C, C++, Java, VHDL, ADA-SPARK, Rust, assembly, bash

Embedded Systems: Arduino, ESP32, Raspberry Pi, FPGA (Basys3), MSP430, PCB Design, GPS/IMU Sensors, RF Communication (2.4 GHz), Communication (SPI, I2C, UART), MIPS, x86, ARM

Tools & Frameworks: Git/GitHub, Docker, Linux (Ubuntu Server), MQTT, Xilinx Vivado, IBM Quantum Composer, Qiskit, CUDA, Fusion360, Formal Verification, 3D-printing, additive manufacturing

AWARDS & ACHIEVEMENTS

1st Place ECE Senior Design Project Award: Open-Source CubeSat Payload (2026)

Outstanding Academic Department; Electrical and Computer Engineering Department GMU (2026)

7th Place National, 1st Place Virginia State, SkillsUSA Additive Manufacturing Competition (2024)

NOVA Design Challenge Winner (2023) | **Fairfax Peace Award** (2022) | **Laura Ashley Piper Award** (2022)

Languages: Arabic (Native), English (Fluent)